

Project Summary Report: Performance Analysis of LLMs and CNNs in Noisy Image Recognition

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Project Background

This project aims to explore methods for enhancing the ability of Large Language Models (LLMs) to handle noisy image recognition [cite: 2]. Using "Four-digit Captcha Recognition" as a specific entry point, the study conducts a comparative analysis between traditional Convolutional Neural Networks (CNNs) and fine-tuned Qwen models to evaluate their performance differences in noisy environments [cite: 2].

I. Project Structure and Core Workflow

The project established a clear hierarchical research structure, covering the full lifecycle from data acquisition to model evaluation [cite: 4]:

- **Data Acquisition:** Utilized multi-threaded crawler scripts to efficiently collect over 100,000 raw captcha images [cite: 5].
- **Data Labeling:** Combined preliminary OCR recognition with high-precision verification via the Kimi API (LLM) to ensure the accuracy of training set labels [cite: 6].
- **Image Preprocessing:** Applied the Otsu binarization algorithm to separate text from complex backgrounds and converted images into .npy matrix formats [cite: 7].
- **Model Construction:**
 - **CNN Branch:** Designed a custom multi-layer convolutional and pooling structure focusing on local feature extraction [cite: 9].
 - **LLM Branch:** Based on Qwen2.5-VL-3B, utilizing LoRA (Low-Rank Adaptation) technology for parameter-efficient fine-tuning [cite: 10].

II. Experimental Data and Performance Comparison

The following table summarizes the performance metrics across different models [cite: 12]:

Evaluation Metric (Accuracy)	CNN (Fine-tuned)	Qwen2.5-VL (Fine-tuned)	Qwen2.5-VL (Zero-shot)
Single Character	85.4%	78.9%	42.5%
Full Image (4-digit)	53.1%	51.2%	24.1%

III. In-depth Project Reflection

1. Data Quality (Garbage In, Garbage Out)

In the early stages of LLM fine-tuning, errors in automated OCR labeling led to model "collapse" (e.g., the model repeatedly outputting gibberish like "ghhd" regardless of the input) [cite: 15]. To rectify this, we initiated 40 concurrent threads to call the Kimi API [cite: 15]. By leveraging Kimi's superior image recognition capabilities, we obtained higher-quality labels, which significantly improved the model's accuracy and performance in recognizing noisy images [cite: 15].

2. Specialized Models (CNN) vs. General Models (LLM)

Results indicate that the CNN slightly outperformed the fine-tuned LLM on this specific task [cite: 17]. Analysis suggests that while LLMs possess exceptional semantic understanding and multi-modal processing capabilities, specialized convolutional kernels are more effective at extracting low-level visual features when dealing with pixel-level noise, severe character overlapping, and geometric distortion [cite: 17]. This demonstrates that "small and precise" fine-tuned models still hold significant value for specific scenarios despite the trend toward "large and all-encompassing" models [cite: 17].

3. Robustness of Preprocessing Strategies

While binarization reduces computational complexity, it may also discard edge details beneficial for recognition [cite: 19]. Future optimizations will consider introducing more data augmentation techniques to allow the model to learn directly from noisy environments rather than relying solely on preprocessing to "filter" noise, thereby enhancing generalization [cite: 19].

IV. Conclusion

This project confirms that LoRA fine-tuning can significantly boost LLM performance in vertical domains, though limitations remain in extreme pixel-level tasks compared to mature CNN solutions [cite: 21]. Future research will focus on combining the low-level perception of CNNs with the high-level reasoning of LLMs to address more complex visual noise challenges [cite: 21].